fn make_bounded_range_to_zCDP

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `fn bounded_range_to_zCDP` in `mod.rs` at commit 0b8f4222 (outdated[1]). This proof is an adaptation of Theorem 5 here, which proves the conversion between bounded range [**?**] and zCDP.

1 Hoare Triple

Preconditions

Compiler-verified

- Variable `eta` must be of type float64

Human-verified

None

Pseudocode

```
def make_bounded_range_to_zCDP(meas: Measurement) -> Measurement:
    def privacy_map(d_in: MI_Distance) -> f64:
        return meas.map(d_in).inf_powi(ibig(2)).inf_div(8.0)

    return meas.with_map(
        meas.input_metric,
        ZeroConcentratedDivergence.default(),
        PrivacyMap.new_fallible(privacy_map),
    )
```

Postcondition

Definition 1.1 (Privacy Loss Random Variable)**.** Let $M : X^n \to \Omega$ be a randomized mechanism, and let $D, D' \in X^n$ be two neighboring datasets. For each possible output $y \in \Omega$, define the *privacy loss at* y as

$$f(y) := \log\left(\frac{\Pr[M(D) = y]}{\Pr[M(D') = y]}\right). \tag{1}$$

Since $M(D)$ is a random outcome in Ω, we can then define the *privacy loss random variable* Z as

$$Z := f(M(D)). \tag{2}$$

In other words, Z is the random variable obtained by first drawing an outcome y from $M(D)$ and then evaluating $f(y)$.

[1] See new changes with `git diff 0b8f4222..002d2dec rust/src/combinators/measure_cast/bounded_range_to_zCDP/mod.rs`

Definition 1.2 (η-Range Bounded)**.** Let $M : X^n \to Y$ be a randomized mechanism and let $\eta > 0$. We say that M is ε-range-bounded if $\forall y \in Y$ and any pair of neighboring datasets $D, D' \in X^n$, there exists a t such that:

$$Z \in [t, t + \eta] \,. \tag{3}$$

Definition 1.3 (Concentrated Differential Privacy)**.** A randomized algorithm $M : \mathcal{X}^n \to \mathcal{Y}$ satisfies ρ-concentrated differential privacy if, for all pairs of inputs $x, x' \in \mathcal{X}^n$ differing only in the data of a single individual and $\forall \lambda > 0$, we have:

$$\mathbb{E}\left[\exp(\lambda Z)\right] \leq \exp\left(\lambda(\lambda + 1)\rho\right), \tag{4}$$

Lemma 1.4. A privacy loss random variable Z satisfies

$$\mathbb{E}\left[\exp(-Z)\right] = 1 \tag{5}$$

Proof.

$$\mathbb{E}\left[(-Z)\right] = \sum_y \mathbb{P}\left[M(x) = y\right] \exp(-f(y)) \tag{6}$$

$$= \sum_y \mathbb{P}\left[M(x) = y\right] \frac{M(x') = y}{M(x) = y} = 1 \tag{7}$$

□

Lemma 1.5. Let M satisfy η-Bounded Range, then it's associated privacy random variable satisfies:

$$\mathbb{E}\left[Z\right] \leq \frac{1}{8}\eta^2 \tag{8}$$

Proof.

$$\mathbb{E}\left[\exp(-Z)\right] \leq \exp\left(-\mathbb{E}[Z] + \frac{\eta^2}{8}\right) \qquad \text{(Via Hoeffding, using } Z \in [t, t+\eta]) \tag{9}$$

$$\implies \mathbb{E}\left[Z\right] \leq \frac{\eta^2}{8} - \log E[\exp(-Z)] \qquad \text{(rearranging the terms)} \tag{10}$$

$$= \frac{\eta^2}{8} \qquad \text{(using Lemma 1.4)} \tag{11}$$

□

Theorem 1.6 (Bounded Range implies zCDP)**.** If M is η-bounded range, then it is $\frac{1}{8}\eta^2$-concentrated differentially private.

Proof.

$$\mathbb{E}\left[\exp(\lambda Z)\right] \leq \exp\left(\mathbb{E}[Z]\lambda + \frac{\eta^2}{8}\lambda^2\right) \qquad \text{(Via Hoeffding, using } Z \in [t, t+\eta]) \tag{12}$$

$$\leq \exp\left(\frac{\eta^2}{8}\lambda + \frac{\eta^2}{8}\lambda^2\right) \qquad \text{(Via Lemma 1.5)} \qquad \leq \exp\left(\frac{\eta^2}{8}\lambda + \frac{\eta^2}{8}\lambda^2\right) \tag{13}$$

$$\leq \exp\left(\frac{\eta^2}{8}\lambda\left(\lambda + 1\right)\right) \tag{14}$$

$$\implies \frac{\eta^2}{8}\text{-zCDP} \qquad \text{(Via Definition 1.3)} \tag{15}$$

□